

Olympiad Test Paper — Social Science (Class 7)

Chapter 3: Climates of India

Subject	Social Science (NCERT Class 7)
Chapter	3 — Climates of India
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Think about the *factor* behind each climate fact — latitude, altitude, distance from the sea, winds or relief — before you choose. Watch the classic traps: confusing weather with climate, thinking the Equator is hot because it is "nearer" the Sun, calling the calm "eye" the most dangerous part of a cyclone, and assuming rain falls equally on both sides of a mountain. Do not write on this paper — mark answers on your answer sheet.

1. Ahaan says, "It is drizzling outside right now." His teacher says, "Cherrapunji is one of the wettest places on Earth." Which pair correctly labels the two statements?

- (A) Ahaan = weather; teacher = climate (B) Ahaan = climate; teacher = weather
(C) Both describe weather (D) Both describe climate

2. A scientist studies thirty years of daily readings for a town and reports the steady, average pattern that emerges. She has described the town's:

- (A) weather (B) climate
(C) season (D) microclimate

3. The single most important difference that separates "weather" from "climate" is the:

- (A) temperature being measured (B) place being described
(C) length of time you look at (D) amount of rainfall

4. Most of the world divides the year into four seasons, but India is usually described as having a fifth. That extra season is the:

- (A) harvest season (B) dry season
(C) windy season (D) rainy season (monsoon)

5. Which of these is NOT one of the six traditional Indian *ṛitus*?

- (A) Vasanta (spring) (B) Varṣhā (rainy)
(C) Maru (desert) (D) Śhīśhir (winter)

6. Seasons return every single year mainly because:

- (A) the Earth revolves around the Sun (B) the monsoon winds reverse
(C) the Sun grows hotter each summer (D) human festivals are repeated yearly

7. India recognises six ritus rather than four seasons chiefly because the six-season scheme:

- (A) is older than the four-season idea (B) was borrowed from European calendars
(C) reads the monsoon-ruled year more finely (D) gives every festival its own season

8. Match the regions to climate types: high Himalayas, Northern Plains, Thar, west coast. The correct order of climate types is:

- (A) alpine, subtropical, arid, tropical wet (B) arid, alpine, tropical wet, subtropical
(C) tropical wet, arid, subtropical, alpine (D) subtropical, alpine, tropical wet, arid

9. India has many different climates at the same time fundamentally because:

- (A) its people follow many different calendars
(B) the Sun shines differently on different states
(C) every state controls its own weather
(D) it is large and varied in latitude, height and nearness to the sea

10. Hill stations like Shimla and Ooty stay cool even in summer. The climate type that best fits these mild lower hills is:

- (A) arid (B) alpine
(C) temperate (D) tropical wet

11. The Northern Plains have very hot summers and cold winters, ideal for wheat. This points to which climate type?

- (A) tropical wet (B) subtropical
(C) alpine (D) maritime

12. Places nearer the Equator are hotter than places near the poles MAINLY because:

- (A) the Equator is far closer to the Sun (B) the poles receive no sunlight at all
(C) there is no atmosphere over the Equator
(D) the Sun's rays strike the Equator almost straight down, concentrating heat

13. Near the poles the Sun's rays arrive slanting. Two things weaken that sunlight: the energy is spread over a wider area, AND the rays:

- (A) reflect off the ice immediately (B) lose heat to passing winds
(C) pass through more air on a longer journey (D) are blocked by mountains

14. Kanniyakumari (near the Equator) is warm all year while Kashmir in the far north is much colder. The climate factor chiefly responsible is:

- (A) altitude (B) distance from the sea
(C) winds (D) latitude

- 15.** The two reasons the chapter gives for temperature falling as you climb a mountain are: the air becomes thinner and less able to hold heat, and:
- (A) you move closer to the cold Sun
 - (B) you move farther from the ground, the air's main heater
 - (C) winds always blow downhill
 - (D) snow reflects all the sunlight
- 16.** Two towns sit at exactly the same latitude. Town A is on a high plateau; Town B is in a low valley nearby. Town A is cooler chiefly because of its greater:
- (A) latitude
 - (B) altitude
 - (C) distance from the sea
 - (D) rainfall
- 17.** The Himalayan peaks stay snow-covered all year while the plains below them swelter. The single factor that best explains this contrast is:
- (A) latitude
 - (B) distance from the sea
 - (C) altitude
 - (D) winds
- 18.** Mumbai (on the coast) has cooler summers and milder winters than Nagpur (far inland), even at nearly the same latitude. This is because:
- (A) Mumbai is at a much higher altitude
 - (B) Nagpur lies much nearer the Equator
 - (C) Nagpur gets no sunshine
 - (D) the sea moderates Mumbai's temperature
- 19.** The sea keeps coastal places mild because water, compared with land:
- (A) heats and cools much more slowly
 - (B) heats and cools much faster
 - (C) reflects all sunlight away
 - (D) never changes temperature at all
- 20.** "Temperature range" means the gap between a place's highest and lowest temperatures. A place with a SMALL temperature range most likely lies:
- (A) near the sea
 - (B) far inland on a high plateau
 - (C) deep inside a desert
 - (D) at a very high latitude
- 21.** Mumbai's yearly temperature range is about 14°C while Nagpur's is much larger. The smaller range at Mumbai is best explained by its:
- (A) higher altitude
 - (B) lower latitude
 - (C) nearness to the moderating sea
 - (D) heavier rainfall
- 22.** The hot, dry "loo" winds that cause summer heat waves in Punjab and Rajasthan get their heat by crossing:
- (A) the Bay of Bengal
 - (B) the snowy Himalayas
 - (C) the Western Ghats
 - (D) the deserts of Arabia and Afghanistan
- 23.** In winter, cold winds slipping over the Himalayas chill the foothills and cause cold waves. This shows that winds affect a region by:
- (A) changing its latitude
 - (B) carrying the qualities of the land they crossed

(C) raising its altitude

(D) moving it nearer the sea

24. The same plain can swelter under a desert wind in one season and shiver under a mountain wind in another. The best explanation is that air:

(A) picks up the character of wherever it has travelled

(B) is created fresh at each place

(C) always warms as it moves

(D) carries no heat or moisture at all

25. The flat Thar suffers hot, dry winds while the lands behind the Himalayas are partly shielded from cold Central-Asian winds. This contrast shows the climate role of:

(A) latitude

(B) topography (relief)

(C) distance from the sea

(D) the monsoon

26. A small shady forest patch stays cooler and moister than the open fields around it. Such a small area with its own special climate is called a:

(A) season

(B) rain shadow

(C) microclimate

(D) ritu

27. "Topography" in this chapter refers to a region's:

(A) population and cities

(B) physical surface features such as mountains, valleys and coasts

(C) yearly festivals

(D) total rainfall only

28. A concrete city centre is several degrees warmer than the green countryside nearby — an "urban heat island". This is essentially an example of a:

(A) season

(B) cyclone

(C) rain shadow

(D) human-made microclimate

29. The word "monsoon" comes from the Arabic *mausim*. What does *mausim* mean?

(A) rain

(B) wind

(C) season

(D) flood

30. Strictly speaking, the word "monsoon" really names the:

(A) rains alone

(B) Indian Ocean itself

(C) six ritus

(D) seasonal winds that reverse with the seasons

31. The whole monsoon system is driven by one basic fact:

(A) the Moon pulls the clouds toward India (B) the Equator stays at a fixed temperature

(C) rivers push the air upward

(D) land heats up and cools down faster than the ocean

32. In summer, moist sea winds rush inland because the hot Asian land has become a zone of:

- (A) high pressure, which pushes air out
- (B) low pressure, into which surrounding air flows
- (C) cold air sinking down
- (D) calm with no pressure difference

33. Air in the monsoon always flows:

- (A) from high pressure to low pressure
- (B) from low pressure to high pressure
- (C) only along the Equator
- (D) straight downward from space

34. The summer monsoon and the winter monsoon are called, in that order, the:

- (A) southwest and northeast monsoons
- (B) northeast and southwest monsoons
- (C) trade-wind and western-disturbance monsoons
- (D) maritime and continental monsoons

35. The summer (southwest) monsoon brings India's main rains because its winds blow:

- (A) from the dry land out toward the sea
- (B) straight down from the upper atmosphere
- (C) from the moist sea onto the land
- (D) only along the Himalayan crest

36. The winter (northeast) monsoon winds are mostly dry because they blow:

- (A) from the moist sea onto the land
- (B) from the land out toward the ocean
- (C) up the Western Ghats
- (D) from the Equator northward

37. The winter monsoon, though mostly dry, still brings some rain to parts of south-east India. It picks up that moisture by crossing the:

- (A) Arabian Sea
- (B) Thar Desert
- (C) Himalayas
- (D) Bay of Bengal

38. The monsoon winds reverse between summer and winter because the:

- (A) Earth changes its tilt twice a year
- (B) land-sea temperature difference flips between the seasons
- (C) Moon changes its orbit
- (D) Western Ghats move with the seasons

39. The monsoon rains usually reach the southern tip of India in early June and cover the whole country by:

- (A) mid-July
- (B) the end of June
- (C) early September
- (D) October

40. The summer monsoon has been compared to "a sea breeze the size of a country" because both work on the same principle:

- (A) the Moon's pull on water
- (B) rivers heating the air above them
- (C) warm land creating low pressure that draws in cooler sea air
- (D) snow melting on the mountains

41. When moist winds are forced up the windward slopes of the Western Ghats, they cool and drop heavy rain. The Deccan Plateau just behind gets far less rain. This dry leeward region is called a:

- (A) microclimate (B) cold desert
(C) rain shadow (D) delta

42. The west coast of India is drenched while the Deccan stays drier mainly because:

- (A) the Deccan is much nearer the Equator (B) the west coast is at a higher altitude
(C) the Deccan has no rivers
(D) the Western Ghats force the monsoon winds up, raining on the western slopes

43. Orographic (relief) rainfall happens because, as air is forced to rise over hills, it:

- (A) warms up and holds more water vapour
(B) expands, cools, and can hold less water vapour, so the vapour condenses
(C) speeds up and dries out instantly (D) sinks back down the windward side

44. If the Western Ghats suddenly vanished, the most likely change to the west coast would be that it:

- (A) received much less rain (B) received much more rain
(C) became a desert overnight (D) had no change at all

45. Suppose land and ocean somehow heated up at exactly the same speed. The monsoon would:

- (A) grow twice as strong (B) reverse only once a year
(C) not exist at all (D) shift to the winter only

46. The place with the highest average annual rainfall in the world, lying in India, is:

- (A) Cherrapunji, Assam (B) Munnar, Kerala
(C) Mumbai, Maharashtra (D) Mawsynram, Meghalaya

47. The "windward" side of a mountain is the side that:

- (A) is sheltered from the wind and stays dry (B) always faces north
(C) receives no sunlight
(D) faces the wind, where moist air rises and rains

48. India's festivals such as Bihu, Pongal, Onam and Baisakhi are described as closely tied to:

- (A) the price of fuel (B) the seasons and farming cycle
(C) the number of cities (D) the latitude of each state

49. "Monsoon failure" means poor or insufficient monsoon rainfall. A likely chain of effects is:

- (A) crops suffer, water grows scarce, and food prices rise
(B) rivers overflow and floods spread everywhere
(C) winters become much warmer (D) the six ěitus disappear

50. Most Indian farming depends on the monsoon, so when the rains fail the economy is hit because:

- (A) festivals are cancelled by law
- (B) the sea stops moderating the coast
- (C) harvests shrink, pushing up food prices and forcing some workers to migrate
- (D) the Himalayas grow taller

51. Every few years the monsoon weakens because warm Pacific water shifts eastward, rearranging tropical winds. This periodic Pacific warming is called:

- (A) the loo
- (B) El Niño
- (C) a glacial burst
- (D) the rain shadow

52. Since the 19th century, climate change has been driven MAINLY by:

- (A) human activities, especially burning fossil fuels
- (B) natural processes alone
- (C) the Earth drifting farther from the Sun
- (D) volcanic eruptions alone

53. The natural greenhouse effect is important for life because greenhouse gases:

- (A) trap some of the Sun's heat, keeping Earth warm enough to live on
- (B) block all of the Sun's rays
- (C) cool the Earth down to freezing
- (D) create the monsoon winds

54. Human-caused global warming happens when burning fossil fuels and clearing forests:

- (A) thin the greenhouse "blanket", letting heat escape
- (B) add extra greenhouse gases, thickening the blanket and trapping more heat
- (C) remove all the air from the atmosphere
- (D) cool the planet by adding smoke

55. Trace the chain "factory burns coal → farmer's monsoon crop fails". The correct order of links is:

- (A) coal burnt → CO₂ released → extra heat trapped → climate shifts → monsoon disrupted → crop fails
- (B) coal burnt → sea cools → monsoon strengthens → crop thrives
- (C) coal burnt → rivers freeze → no irrigation → crop fails
- (D) coal burnt → forests grow → more rain → crop floods

56. Which of these is an example of climate "mitigation"?

- (A) cutting down more forests
- (B) burning more coal for cheap power
- (C) increasing wasteful consumption
- (D) boosting renewable energy and planting trees

57. Mitigation and resilience attack climate change in two different ways. The best summary is:

- (A) mitigation slows the warming; resilience helps communities withstand and recover from impacts
- (B) mitigation builds flood walls; resilience cuts emissions

(C) both only plant trees

(D) mitigation ignores the cause; resilience ignores the effects

58. A cyclone forms over the sea when the air pressure there becomes:

(A) much higher than its surroundings, pushing air out

(B) colder than the land in winter (C) exactly equal everywhere

(D) much lower than its surroundings, so air and moisture rush in

59. During a cyclone there is suddenly an eerie calm with clear sky overhead. The townspeople should:

(A) assume the storm is over and go outside (B) head to the coast to watch

(C) stay sheltered, because the calm "eye" has passed over and fierce winds will return

(D) light a fire to dry out

60. The fiercest winds of a cyclone are found:

(A) in the calm, cloudless "eye" at the centre (B) in the ring of cloud surrounding the eye

(C) hundreds of kilometres out at sea (D) only after the cyclone has ended

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Social Science (Class 7), Chapter 3: Climates of India

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	B	21	C	31	D	41	C	51	B
2	B	12	D	22	D	32	B	42	D	52	A
3	C	13	C	23	B	33	A	43	B	53	A
4	D	14	D	24	A	34	A	44	A	54	B
5	C	15	B	25	B	35	C	45	C	55	A
6	A	16	B	26	C	36	B	46	D	56	D
7	C	17	C	27	B	37	D	47	D	57	A
8	A	18	D	28	D	38	B	48	B	58	D
9	D	19	A	29	C	39	A	49	A	59	C
10	C	20	A	30	D	40	C	50	C	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

- (A)** "Drizzling right now" is what is happening this moment — that is weather. "Cherrapunji is one of the wettest places on Earth" describes a long-term pattern over decades — that is climate. So Ahaan = weather, teacher = climate. The trap (B) reverses the two; the catch is that a single moment is always weather, never climate.
- (B)** Averaging weather over thirty years to find the steady pattern is the very definition of climate. Weather (A) is hour-to-hour; a season (C) is a yearly recurrence; a microclimate (D) is a small local area — none fits a 30-year average for a whole town.
- (C)** Weather and climate describe the *same* atmosphere and can describe the same place and rainfall — the ONLY thing that separates them is the length of time you look at: a moment is weather, decades give climate. Hence (C), not (A), (B) or (D).
- (D)** Most of the world uses four seasons; India is famous for adding the rainy season, the monsoon, as a fifth. "Harvest", "dry" and "windy" seasons (A, B, C) are not the chapter's named extra season.

- 5. (C)** The six *ṛitus* are *vasanta*, *grīṣhma*, *varṣhā*, *śharad*, *hemanta* and *śhiśhir*. "Maru" (desert) is not a *ṛitu* — it is a distractor built from the Thar. *Vasanta*, *Varṣhā* and *Śhiśhir* are all genuine *ṛitus*, so the odd one out is (C).
- 6. (A)** Seasons recur every year because the Earth revolves around the Sun, so the same conditions return annually. Reversing monsoon winds (B) explain the two monsoons, not the yearly return of all seasons; the Sun does not grow hotter each summer (C); festivals (D) follow the seasons, they don't cause them.
- 7. (C)** India uses six *ṛitus* because that finer scheme reads a monsoon-ruled year better than four seasons — a four-season model has no slot for the all-important rains. Age (A), borrowing from Europe (B) and festival-matching (D) are not the chapter's reason.
- 8. (A)** High Himalayas = alpine, Northern Plains = subtropical, Thar = arid, west coast = tropical wet. Only (A) lists them in that order; the other options scramble the pairing.
- 9. (D)** A country stretching from snowy mountains to a hot desert to rainy coasts cannot have one climate: its range of latitude, altitude and nearness to the sea produces many climates side by side. Calendars (A), a "different Sun" (B) or states "controlling weather" (C) are not real causes.
- 10. (C)** Lower hills with hill stations that stay mild are temperate. Arid (A) is the dry Thar; alpine (B) is the freezing high Himalayas; tropical wet (D) is the rainy west coast — none describes mild lower hills.
- 11. (B)** Very hot summers, cold winters and good wheat country describe the subtropical Northern Plains. Tropical wet (A) is the rainy coast; alpine (C) is freezing mountains; maritime (D) means a mild coastal climate — the opposite of "hot summers and cold winters".
- 12. (D)** The Equator is hotter mainly because the Sun's rays strike it almost straight down, concentrating their heat on a small area. The "closer to the Sun" idea (A) is the classic misconception — the distance difference is tiny. There is air over the Equator (C) and the poles do get some sunlight (B); both are false.
- 13. (C)** Slanting polar rays are weakened two ways: spread over a wider area AND passing through more air on a longer journey. Reflecting off ice (A), losing heat to winds (B) and being blocked by mountains (D) are not the chapter's two reasons.
- 14. (D)** Kanniyakumari is warm and Kashmir cold because of how far each lies from the Equator — that is latitude. Altitude (A) would matter for a mountain-vs-plain pair, distance from the sea (B) for a coast-vs-inland pair, winds (C) for a seasonal contrast — but the equator-to-far-north comparison is a latitude effect.

15. (B) The two reasons given are: the air higher up is thinner and holds less heat, AND you are farther from the ground, which is the air's main heater. You do not move "closer to the cold Sun" (A — the Sun isn't cold and you barely move toward it); (C) and (D) are invented.

16. (B) Same latitude rules out latitude; the high plateau town is cooler because of its greater altitude — temperature falls as you climb. Distance from the sea (C) and rainfall (D) are not what sets these two apart.

17. (C) Peaks are snowbound while the plains below swelter — the towering height of the peaks (altitude) is the deciding factor, since temperature drops with elevation. Latitude (A) is similar for peak and plain; sea distance (B) and winds (D) don't explain the snow line.

18. (D) Same latitude, so latitude is ruled out; Mumbai is milder because the sea moderates its temperature (water warms and cools slowly). "Higher altitude" (A) is false for coastal Mumbai; Nagpur is not nearer the Equator (B); it certainly gets sunshine (C).

19. (A) The sea moderates because water heats and cools much more slowly than land, storing summer heat and releasing it gently in winter. (B) is the reverse and would make coasts extreme; (C) and (D) are false — the sea does change temperature, just slowly.

20. (A) A small temperature range means an even, moderate climate — typical of places near the sea, which the ocean keeps mild. A high inland plateau (B), a desert (C) and a very high latitude (D) all give large ranges (big extremes), the opposite of small.

21. (C) Mumbai's small $\sim 14^{\circ}\text{C}$ range is explained by its nearness to the moderating sea, which evens out summer and winter. Altitude (A) is low at the coast; lower latitude (B) would affect overall warmth, not the smallness of the range; rainfall (D) does not set the temperature range.

22. (D) The loo gets its heat by crossing the hot deserts of Arabia and Afghanistan before reaching north India. The Bay of Bengal (A) and the Western Ghats (C) would add moisture, not dry heat; the snowy Himalayas (B) would chill, not heat, the wind.

23. (B) Cold Himalayan winds chilling the foothills show that winds carry the qualities (here, cold) of the land they have crossed. Winds do not change a place's latitude (A) or altitude (C), nor move it nearer the sea (D).

24. (A) The same plain can be hot under a desert wind and cold under a mountain wind because air picks up the character of wherever it has travelled — heat from deserts, cold from snowy mountains. Air is not "created fresh" (B), does not always warm as it moves (C), and certainly carries heat and moisture (D).

25. (B) Flat Thar lying open to hot winds while ranges shield the land behind them shows the climate role of topography (relief) — the shape of the land blocking or admitting winds.

It is not about latitude (A), distance from the sea (C) or the monsoon (D).

26. (C) A small patch with its own special climate is a microclimate. A season (A) and a ritu (D) are time periods, not places; a rain shadow (B) is the dry leeward side of a mountain, not a shady forest patch.

27. (B) Topography means a region's physical surface features — mountains, hills, valleys, slopes and coasts. It is not population/cities (A), festivals (C) or rainfall alone (D).

28. (D) A concrete city warmer than the green countryside is a human-made microclimate (an urban heat island). It is not a season (A), a cyclone (B) or a rain shadow (C).

29. (C) *Mausim* (Arabic) means "season" — which is why "monsoon" really refers to seasonal winds. Rain (A), wind (B) and flood (D) are tempting because we link the monsoon to rains and winds, but the literal meaning is "season".

30. (D) Strictly, "monsoon" names the seasonal winds that reverse direction with the seasons; people only loosely use it to mean the rains (A) those winds bring. It is not the six ritus (C) or the ocean itself (B).

31. (D) The whole monsoon turns on one fact: land heats and cools faster than the ocean, creating the pressure differences that move the winds. The Moon (A), a "fixed" Equator temperature (B) and rivers pushing air (C) are all false.

32. (B) In summer the hot land becomes a low-pressure zone; since air flows from high to low pressure, moist sea air rushes inland. If the land were high pressure (A) the air would flow out (that is winter); (C) and (D) describe no inflow.

33. (A) Air always flows from high pressure to low pressure — that is what drives the monsoon winds. Option (B) is backwards; (C) and (D) are false.

34. (A) The summer monsoon is the southwest monsoon and the winter monsoon is the northeast monsoon — in that order, (A). Option (B) reverses them; (C) and (D) invent names.

35. (C) The summer monsoon brings the main rains because its winds blow from the moist sea onto the land, carrying water vapour that falls as rain. Land-to-sea winds (A) are the dry winter monsoon; (B) and (D) are false.

36. (B) The winter (northeast) monsoon is mostly dry because its winds blow from the land out toward the ocean, carrying little moisture. Sea-to-land winds (A) are the wet summer monsoon; (C) and (D) are false.

37. (D) Part of the winter monsoon crosses the Bay of Bengal, picking up moisture and bringing some rain to south-east India. The Arabian Sea (A) feeds the summer monsoon's west coast; the Thar (B) and Himalayas (C) would not add moisture to these winds.

38. (B) The winds reverse because the land–sea temperature difference flips between summer and winter, flipping which side is high pressure. The Earth's tilt (A) and the Moon's orbit (C) don't reverse twice a year in this way, and the Ghats (D) do not move.

39. (A) The rains reach the southern tip in early June and cover the whole country by mid-July. End of June (B) is too soon; September (C) and October (D) are too late.

40. (C) Both a sea breeze and the summer monsoon work by warm land creating low pressure that draws in cooler sea air — the monsoon is just that breeze on a country-sized scale. The Moon (A), rivers (B) and melting snow (D) are not the principle.

41. (C) The dry leeward region behind the Western Ghats, left in the "shadow" of the rain, is a rain shadow. A microclimate (A) is a small local area, a cold desert (B) is a high-altitude dry zone, and a delta (D) is a river mouth — none names this dry plateau.

42. (D) The west coast is wet because the Western Ghats force the moist monsoon winds to rise and rain on the western slopes, leaving the Deccan drier. The Deccan is not nearer the Equator (A); the west coast is not higher (B); river count (C) does not decide rainfall.

43. (B) Orographic rain forms because rising air expands, cools, and can hold less water vapour, so the vapour condenses into rain. Rising air does not warm or hold more vapour (A); it does not dry instantly (C) or sink on the windward side (D).

44. (A) Without the Ghats there would be nothing to force the moist winds up, so the west coast would receive much less rain. "Much more" (B) and "more rain → flood" reasoning are backwards; "desert overnight" (C) overstates it; "no change" (D) ignores the Ghats' role.

45. (C) The monsoon exists only because land heats and cools faster than the ocean. If both heated at the same speed, there would be no pressure difference and no monsoon at all. It would not strengthen (A), reverse once (B) or shift to winter (D).

46. (D) Mawsynram in Meghalaya receives the highest average annual rainfall in the world. Cherrapunji (A) is nearby and very wet but not the highest; Mumbai (C) and Munnar (B) are wet but far below Mawsynram.

47. (D) The windward side faces the wind, where moist air is forced up, cools and drops heavy rain. The sheltered, dry side (A) is the leeward side; (B) and (C) are unrelated to the term.

48. (B) Festivals like Bihu, Pongal, Onam and Baisakhi are tied to the seasons and the farming cycle. They are not tied to fuel prices (A), the number of cities (C) or each state's latitude (D).

49. (A) Monsoon failure means poor rains, so crops suffer, water grows scarce and food prices rise. Overflowing rivers and floods (B) come from too much rain, not failure; warmer winters (C) and vanishing ritus (D) are not its effects.

50. (C) Because most farming depends on the monsoon, failed rains shrink harvests, push up food prices and force some workers to migrate to cities. Festivals are not "cancelled by law" (A); the sea keeps moderating the coast (B); the Himalayas do not grow (D).

51. (B) The periodic eastward shift of warm Pacific water that weakens India's monsoon is El Niño. The loo (A) is a hot land wind, a glacial burst (C) is a Himalayan flood, and a rain shadow (D) is a dry mountain region.

52. (A) Since the 1800s, climate change has been driven mainly by human activities, especially burning fossil fuels (plus deforestation). "Natural processes alone" (B) was true long ago, not since the 19th century; (C) and (D) are not the main driver.

53. (A) The natural greenhouse effect matters because greenhouse gases trap some of the Sun's heat, keeping Earth warm enough for life. They do not block all rays (B) or cool Earth to freezing (C), and they do not create the monsoon winds (D).

54. (B) Human warming happens when burning fossil fuels and clearing forests add extra greenhouse gases, thickening the heat-trapping "blanket" so more heat is trapped. Thinning the blanket (A) would cool, not warm; (C) and (D) are false.

55. (A) The cause-effect chain runs: coal burnt → CO₂ released → extra heat trapped (greenhouse effect) → climate shifts → monsoon disrupted → crop fails. Burning coal does not cool the sea (B), freeze rivers (C) or grow forests (D), so only (A) is a valid chain.

56. (D) Mitigation attacks the *cause* of warming — boosting renewable energy and planting trees both cut or absorb greenhouse gases. Cutting forests (A), burning more coal (B) and wasteful consumption (C) all worsen warming, the opposite of mitigation.

57. (A) Mitigation slows the warming by attacking its cause (cutting emissions, adding trees); resilience helps communities withstand and recover from impacts like floods. Option (B) swaps the two; (C) oversimplifies; (D) is false — each addresses one side, not neither.

58. (D) A cyclone forms when the sea-air pressure drops much lower than its surroundings, so air and moisture rush in and spiral up. High pressure (A) pushes air out, equal pressure (C) gives no storm, and being "colder than the land" (B) is not what triggers a cyclone.

59. (C) The eerie calm with clear sky is the cyclone's "eye" passing overhead; the fierce winds of the surrounding ring will return, so people must stay sheltered. Going outside (A) or to the coast (B) is dangerous; lighting a fire (D) is irrelevant.

60. (B) The fiercest winds are in the ring of cloud surrounding the eye, not in the calm "eye" itself (A) — the common misconception. They are not far out at sea (C) or only after the storm ends (D).

Solutions complete: 60 / 60.